INNOVATION-ORIENTED MANAGEMENT OF THE IMPLEMENTATION OF SUSTAINABLE DEVELOPMENT GOALS

Yuliia Papizh
Candidate of Economic Sciences, Associate Professor,
Dnipro University of Technology, Dnipro, Ukraine

The article analyzes the latest data from the Copernicus Climate Change Observatory, which operates under the management of the European Commission, on exceptionally high air temperature anomalies in June 2025 in most of Europe. The article states that, under the existing conditions, in the interests of sustainable development of civilization, the imperative strategy for the development of the global economy is a decarbonization strategy, which involves growth using the broad innovation capabilities of the world's leading countries, their economic integration. In this regard, the need for targeted development of innovative technologies and investments is emphasized, taking into account the potential of universities and laboratories in Europe, the USA, Japan, other countries of the world, their financial resources for the implementation, in particular, of global projects in the following areas: research into thermonuclear energy, electromobility, development of promising battery technologies, wind and solar energy, use of hydropower. For the implementation of international innovative, the issues of strategic management effectiveness are of particular importance.

Keywords: strategic management, economic integration, innovation, technology, decarbonization, globalization, efficiency, sustainable development.

* * * *

Climatic changes have affected the entire planet, but their negative manifestations are especially acute in Europe, which is historically the most important region, where modern human civilization was largely formed, which continues to be the most important economic and innovative center of the planet, determining the nature of the world economy.

Extreme climate change of the planet requires adequate measures of the entire world community to decarbonize the economy. The strategy of decarbonization and effective strategic management must be based (Fig. 1). on the pursuit of the goals of sustainable development of the United Nations, and the global nature and extremely great importance of the tasks for modern civilization and future generations, as well as significant experience and obvious expediency, imply broad international cooperation, economic integration, and the use of innovative and investment potentials of the world's leading countries.

Figure. 1. Decarbonization strategy and its elements

The most important tool of strategic management in the conditions of parliamentary democracy is the support of decarbonization projects by the population of countries, voters who form the government and influence decision-making. Without the support of voters, sustainable financing of many projects, especially large-scale ones that do not lead to a quick positive result, may end. The task of specialists, scientists and experts, managers of various levels are to fully and reliably inform the broad circles of the international public about the processes of climate change and the relevance of decarbonization in the interests of sustainable development.

Analysis of the development of the thermonuclear project showed the following.

1. The international community understands the colossal importance of the thermonuclear project for providing human civilization with an almost inexhaustible source of energy and is fully determined to continue its active development with the participation of the world's leading countries.

2. The thermonuclear project is probably the most complex technological innovation project in the history of mankind and potentially the most large-scale in terms of economic significance, in case of its successful implementation.

3. There is no doubt that the activity of ITER will have the most important scientific significance, but the question remains whether it will lead to the expected practical results.

4. Already at the present time, as of mid- 2025, it is quite obvious that the initial project implementation deadlines cannot be met, with a high probability it can be assumed that the new, upwardly adjusted project implementation deadlines will not be met and its com-

pletion will be delayed in time, but it is impossible to predict exactly how much.

5. The problems of increasing the cost of the project are already noticeable and, very likely, will only increase.

6. Most likely, the main participants of the project will continue large-scale investment and innovation efforts for its development, despite the emergence of many problems that will accompany the project, and it will be implemented.

7. In our opinion, despite the extreme complexity, the goals of the project do not cause doubts and it should be continued, increasing the potential of international cooperation, investments and innovations even in conditions when the solution to the problem of the development of thermonuclear power plants is not obvious.

8. With a high probability, the completion time and the cost of the project will be significantly increased, which is caused by its colossal scale and complexity. In these conditions, long-term financing of the project is most likely possible only with broad public support of the voters of democratic countries and the authorities elected by them, which requires constant informing of broad circles of the public about the extreme importance and necessity of the project, which is a key tool of strategic management.

Among the mass modes of transport on a global scale, electric vehicle transport can be called the most strategically promising. The most important advantage of electric cars is their climate neutrality, which corresponds to the global processes of decarbonization and sustainable development. Individual electric cars are universal – no other type of transport for trips over relatively long distances can fully replace electric cars. Public transport runs on a schedule that may not meet people's needs. In addition, public transport routes do not cover significant geographical areas, the use of public transport requires transfers, waiting for dispatch, and often delays in the arrival of routes. All these factors create specific problems for the use of public transport for many people. Individual transport in the form of electric cars is free from such shortcomings.

The cost of electric energy when operating electric cars is significantly lower than the similar fuel costs of traditional cars. In different cases, there may be a several-fold difference in cost. Here, it is important to take into account additional current circumstances that can significantly affect the comparative efficiency of electric cars. Modern trends in the world oil market are very complex and contradictory in nature, but in general they are characterized by the following factors: high world oil prices and the general direction of their increase, even in spite of noticeable price fluctuations in certain periods. The trend of the general growth of world oil prices is connected with the noticeable growth of the planet's population, connected with this increase in oil consumption, the growth of oil consumption per capita, the reduction of available oil reserves in the world, which is inevitably accelerating due to the colossal annual production volumes. In these conditions, the economic benefits of using electric cars will only increase. This increase will be additionally supported by the dominant trend of a noticeable decrease in the cost of electric car batteries and the cost of electric cars themselves.

In the modern conditions of decarbonization, the fact that the owners of electric cars can quite reasonably experience a sense of involvement in the progressive forces of society in terms of innovative factors of sustainable development and participation in the multifaceted process of preserving the planet's climate, especially in the conditions of global warming, which is becoming more and more pronounced and causing great economic damage, is of key additional importance in modern conditions of decarbonization.

A negative and very significant factor, which probably holds back the global development of electric mobility to the greatest extent, is that the cost of electric cars at the present time still significantly exceeds the cost of traditional cars, which is primarily due to the high cost of the most important element of electric cars - batteries. The capacity of batteries in most cases is such that it does not allow them to make a mileage similar to the mileage of traditional cars on one charge. But the situation in the direction of increasing the capacity of batteries is constantly changing under the influence of innovations. However, in recent years, the cost of accumulator batteries and their energy capacity have decreased many times.

The problems of the development of electric mobility also include the insufficient number and capacity of chargers, the long charging time, which significantly complicates the use of electric cars, especially when traveling long distances.

The strategy for the development of electric mobility should provide for the gradual complete replacement of traditional cars with internal combustion engines by electric cars, which represents the largest technological revolution of the 21st century. Such a strategy envisages a set of measures for the mass production of electric cars, the corresponding infrastructure, the most important tasks of the development strategy and innovative management are the development of new designs of batteries with a significantly higher energy capacity and lower cost, including through the use of new materials, acceleration of battery charging, development and production of electric cars with the lowest possible cost, but acceptable technical and other indicators that will allow such electric cars to be widely distributed on the world market.

The number of international project participants and the complexity of the tasks emphasize the importance of effective strategic management for the successful implementation of projects and clearly shows the effectiveness of the development of the global economy.

Thus, the study showed that the efforts of the world community in the direction of decarbonization of the world economy in the interests of sustainable development of civilization are in the center of attention of the economically leading countries of the world. Such efforts do not currently allow to stop the negative changes in the planet's climate. Effective strategic management and joint innovative projects of many countries, universities, research centers, scientific societies, especially Europe, the USA, and Japan in priority directions can be used as the most important tool for decarbonization.

Successful development of priority innovative technologies requires effective strategic management and global efforts in the interests of sustainable development.

REFERENCES:

1. Bilous S. P, Trokhymenko A. S., Kaminskyi V. V. (2024). Stratehichne upravlinnia rozvytkom pidpryiemstva v umovakh kryzy ta viiskovykh vyklykiv [Strategic management of enterprise development in times of crisis and military challenges]. *Ekonomika ta suspilstvo*, vol. 68. Available at: https://economyandsociety.in.ua/index.php/ journal/article/view/4935/4879

2. Dudnyk O. V. (2025). Mekhanizmy formuvannia stratehii rozvytku pidpryiemstv ahropromyslovoho kompleksu [Mechanisms for forming development strategies for agro-industrial enterprises]. *Ekonomichnyi prostir*, vol. 201. Available at: https://economic-prostir.com.ua/wp-content/uploads/2025/06/201-61-66-dudnyk.pdf

3. Borysiak O., Mucha-Kuś K., Brych V., Kinelski G. (2022). Toward the Climate-Neutral Management of Innovation and Energy Security in Smart World: monograph. Berlin, Germany: Logos Verlag Berlin GmbH.

4. Couckuyt D., Van Looy A. (2020). A systematic review of Green Business Process Management. *Business Process Management Journal,* Vol. 26, no. 2, pp. 421–446. DOI: https://doi.org/10.1108/BPMJ-03-2019-0106

5. Gouda K.C., Thirumalai Raja R. (2022). The benefits of a net-zero emission policy for the environment, energy security, and energy equity (3Es) in developing nations. *Journal of Nonlinear Analysis and Optimization*. Vol. 13. Issue 2, no. 2. Available at: https://www.jnao-nu.com/Vol.%2013,%20Issue.%2002,%20July-December%20:%202022/45.2.pdf

6. Sohns T.M., Aysolmaz B., Figge L., & Joshi A. (2023). Green business process management for business sustainability: A case study of manufacturing small and medium-sized enterprises (SMEs) from Germany. *Journal of Cleaner Production*. DOI: https://doi.org/10.1016/j.jclepro.2023.136667

7. Shu Zhang, Yubo Ma, Xinzhu Zheng, Qianting Zhu, Xu Tang (2024). Reconstruction and Implications of Energy Security Assessment System in the Context of Green and Low-Carbon Transition. *Strategic Study of CAE*, vol. 26. Issue (4), pp. 28–39. DOI https://doi.org/10.15302/J-SSCAE-2024.04.016

8. Copernicus: Heatwaves contribute to the warmest June on record in western Europe. (2025). Available at: https://climate.copernicus.eu/heatwaves-contribute-warmest-june-record-western-europe

9. ITER Members. (2025). Available at: https://www.iter.org/about/iter-members

10. Building ITER. (2025). Available at: https://www.iter.org/building-iter

11. Summary of Presentation by Pietro Barabaschi, ITER Director-General. (2024). Available at: https://www.iter.org/sites/default/files/media/2024-7/baseline_press_conference_summary_july-2024_b.pdf/